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Limited influence of stream networks on the terrestrial movements of three wetland-dependent frog species

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Abstract

Quantifying functional connectivity is essential for understanding factors that limit or promote animal dispersal in fragmented landscapes. Topography is a major factor influencing the movement behavior of many animal species, and therefore the extent of functional connectivity between habitat patches. For pond-breeding frogs, areas of low topographic relief (such as streams or drainage lines) offer damp microhabitats that can facilitate movement through otherwise dry landscapes. However, the extent of topographic bias of frog movements has rarely been quantified. We used a replicated study to compare captures in high- and low-relief transects, for three species from a pond-breeding frog community in southeastern Australia. We captured frogs significantly more often on low-relief transects. However, capture rates decreased with increasing distance from water at similar rates on both high-relief and low-relief transects, and we observed few differences between adult and juvenile movements. Our results suggest that although low-relief drainage lines are important for the pond-breeding frogs in question, ecologists and landscape managers should not discount the role of high-relief locations. Because low-relief drainage lines represent a low proportion of the pond margin, >90% of movements are likely to occur across high-relief locations. Therefore, for the species that we studied, buffer zones designed to conserve only

hydrological networks would provide insufficient protection of frequently used pond margins, while drainage lines are unlikely to act as vital networks facilitating connectivity between breeding ponds. Our study suggests that movement across slopes may be most important for facilitating functional connectivity.

Highlights

► Topography strongly influences functional connectivity for many animal taxa. ► For frogs, the extent of topographic movement bias is a major research gap. ► We used a replicated trapping design to quantify this bias. ► High-relief areas had low frog densities, but contained more individuals in total. ► Buffering hydrological networks is insufficient to conserve pond-breeding frogs.

Introduction

Habitat fragmentation, caused by large-scale human modification of ecosystems, is a major driver of biodiversity loss (Fahrig, 2003, Kingsford et al., 2009, Lindenmayer and Fischer, 2006). Conceptual landscape models which emphasize patch-matrix habitat distributions (derived from Island Biogeography Theory; MacArthur and Wilson, 1967) can be useful for describing fragmented landscapes, and have therefore been influential throughout the development of landscape ecology (Haila, 2002). Applications of these models commonly assume that viable metapopulations are maintained by dispersal (Hanski, 1998, Leibold et al., 2004), while acknowledging that dispersal can be strongly influenced by properties of the intervening matrix (Vogt et al., 2009). Quantifying the extent to which parts of the landscape facilitate animal movement – a concept known as 'functional connectivity' (Baguette and Van Dyck, 2007, Lindenmayer and Fischer, 2007) – provides a basis for understanding the effects of matrix alteration on patch-dependent animal populations (see Storfer et al., 2010).

Functional connectivity is a particularly relevant concept for frog populations. Pond-breeding frogs are commonly described as a naturally occurring model of a fragmented system, because ponds appear like patches in a terrestrial matrix (Bradford et al., 2003, Marsh and Trenham, 2001). For this reason, metapopulation theory (Hanski, 1998) has commonly been used as a model for describing frog populations (Smith and Green, 2005). However, dispersal rates are highly variable between frog species (e.g. Driscoll, 1997, Smith and Green, 2006), and the role of landscape resistance in explaining this variation remains unclear (Stevens et al., 2006). In particular, topographic features represent barriers to movement in some species and locations (e.g. Funk et al., 2005, Richards-Zawacki, 2009, Richter-Boix et al., 2007) but not others (e.g. Davis and Roberts, 2005, Driscoll, 1998, Zhan et al., 2009). Functional connectivity therefore provides a framework for investigating the influence of landscape variation on frog dispersal and for deciding, in turn, which management interventions are likely to be effective for conservation (see Petranka and Holbrook, 2006).

Although there has been much research into terrestrial habitat use by frogs (Baldwin and deMaynadier, 2009, Bulger et al., 2003, Parker and Anderson, 2003, Patrick et al., 2008, Semlitsch and Bodie, 2003), relatively little work has focused on the concept of functional connectivity at fine spatial scales (although see e.g. Popescu and Hunter, 2011, Stevens et al., 2006, Todd et al., 2009). This is unusual given that conditions in the pond margin can strongly influence both landscape resistance (Semlitsch et al., 2009) and emigration orientation (Mazerolle and Vos, 2006, Timm et al., 2007b), thereby inducing large differences in functional connectivity between patches at landscape scales. It is important, therefore, that investigations of landscape resistance for frogs include research into behavior at the pond margin.

In this paper, we describe a study designed to address the question: Do frogs preferentially use areas of low topographic relief within pond margins? High-relief movement paths require more energy to cross than low relief paths (Lowe et al., 2006), and contain proportionally fewer damp microhabitats that provide refugia from desiccation (Rittenhouse et al., 2009). Further, there is evidence both that some species rely on drainage-lines to facilitate terrestrial movements (Rittenhouse and Semlitsch, 2007), but also that overland dispersal can be an important process facilitating species persistence in some cases (Grant et al., 2010, Hazell et al., 2001). These examples suggest that the role of topographically defined barriers and movement corridors warrants further attention in relation to functional connectivity for frogs. However, the influence of topography on frog movements has received proportionally less attention than factors such as vegetation structure (e.g. see Semlitsch et al., 2009 and references therein).

We used a replicated, trap-based approach to quantify fine-scale variation in frog movement behavior, taking into account several sources of variation including the effects of rainfall, migration, demography, and distance from water on capture rates, as well as topography. Our guiding assumption was that the need to avoid desiccation is an important mechanism driving spatial and temporal variability in frog terrestrial movements. Consequently, we anticipated that captures in relation to topographic relief would be influenced by both distance from water and rainfall.

Insights into the influence of topography on frog movements are important because they have practical implications for conservation efforts. In particular, frog species with a high proportion of hydrological network-biased movement will be effectively conserved using buffer zones surrounding streams and breeding ponds (see Semlitsch and Bodie, 2003), while species which predominantly display overland movements will not. More generally, our study provides a direct, replicable test of landscape resistance. Such studies are rare, but are fundamental to understanding and managing connectivity in fragmented landscapes (Fahrig, 2007).

Section snippets

Study area

Our study area was Booderee National Park, in the Jervis Bay Territory, south-eastern Australia (approximate coordinates 35°10′S 150°40′E; see Fig. 1). The park covers the majority of the southern peninsula of Jervis Bay. It is owned by the Wreck Bay Aboriginal Community, and co-managed in association with the Australian Department of Sustainability, Environment, Water, Population and Communities (SEWPaC). The study region has a temperate climate, with average annual rainfall of approximately …

Results

We captured a total of 965 frogs from seven species: 538 in our spring trapping period, and 427 in our summer trapping period. However, during our spring trapping period, three species – *Pseudophryne bibronii*, *Litoria jervisiensis* and *Heleioporus australiacus* – were represented by only a single individual, and we captured only four adults from a fourth species (*Limnodynastes peronii*). This left three species that were sufficiently common to enable us to construct models of adult occurrence from …

Discussion

For this study, we aimed to quantify the extent of topographic bias in the movement behavior of three frog species. We found that low-relief drainage lines providing the majority of inflow into breeding ponds were preferentially used by these frogs. More specifically, we found statistically significant differences in frog occurrence between high- and low-relief transects for all three species, although this effect was mediated by rainfall and distance to water (Fig. 2). While this would appear …

Conclusions

Our study has shown that – for the pond-breeding frogs that we examined – the majority of individuals used terrestrial areas that were not located on drainage lines. This leads to two important conclusions for frog conservation in this location. First, buffer zones designed to conserve only hydrological networks would provide insufficient protection of locations that were commonly used by frogs in this study. Second, drainage lines are unlikely to be vital networks facilitating connectivity …

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